



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

UNIT TEST

23-05-2021

JEE(Main + Advance) : Enthusiast (PHASE : TAAS,TRAS,TEAS,TTAS,TOAS,TNAS,TMAS)

Test Type : UNIT TEST

Test Pattern : JEE (Advanced)

TEST DATE : 23-05-2021

ANSWER KEY

PHYSICS

SEC-1	Q.	1	2	3	4				
	A.	C	D	D	A				
SEC-2	Q.	1	2	3	4	5	6	7	8
	A.	A,B,C,D	A,C	B	A,C	C,D	A,B,C	A,C	A,B,C
SEC-3	Q.	1	2	3	4	5	6		
	A.	3.00	60.00	4.75 to 4.80	11.00	5.00	4.05 to 4.35		

CHEMISTRY

SEC-1	Q.	1	2	3	4				
	A.	A	B	B	D				
SEC-2	Q.	1	2	3	4	5	6	7	8
	A.	A,B,C,D	A,D	A,B,C	A,B,D	A,B,D	A,C,D	A,B,C,D	A,C,D
SEC-3	Q.	1	2	3	4	5	6		
	A.	54.44	45.00	5.00	14.00	2.00	7.00		

MATHEMATICS

SEC-1	Q.	1	2	3	4				
	A.	B	C	B	A				
SEC-2	Q.	1	2	3	4	5	6	7	8
	A.	B	A,B,C,D	A,B	A,C	C	B,C	B,C,D	B,D
SEC-3	Q.	1	2	3	4	5	6		
	A.	1.75	7.50	0.25	0.40	0.00	0.00		

HINT – SHEET

PHYSICS

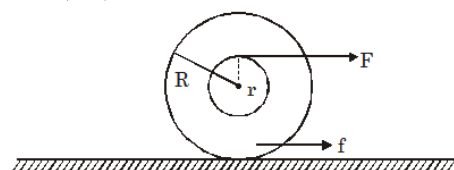
SEC-1

1. Ans (C)

The circuit can be reduced as parallel combination of two cubes having resistance of each side as R and net resistance is found about body diagonal

$$\therefore R_{eq} = \left(\frac{5}{6}\right) \times \frac{1}{2} = \frac{5}{12} \Omega$$

2. Ans (D)



$$F + f = Ma \dots\dots(1)$$

$$Fr - fR = I\alpha$$

$$= MRr \left(\frac{a}{R}\right)$$

$$\Rightarrow Fr - fR = Mar \dots\dots(2)$$

$$\text{from (1) } f = Ma - F$$

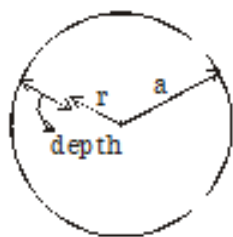
$$\Rightarrow Fr - (Ma - F)R = Mar$$

$$\Rightarrow F = Ma \dots\dots(3)$$

$$\text{from (1) and (3) } f = 0$$

As the friction required is zero, therefore sphere will roll without slipping for every value of force.

3. Ans (D)



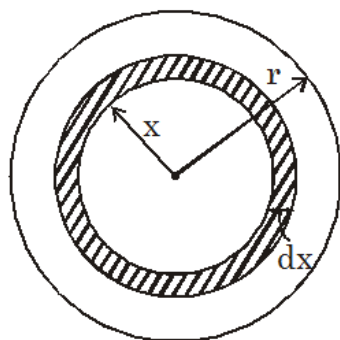
depth at distance $r = a - r$

$$\therefore \rho(r) = \rho_0(a - r)$$

$$\int \vec{E} \cdot d\vec{A} = \frac{q_{in}}{\epsilon_0}$$

$$E(4\pi r^2) = \frac{q_{in}}{\epsilon_0}$$

Now, we will calculate q_{in}



$$dv = 4\pi x^2 dx$$

$$dq = \rho dv$$

$$q_{in} = \int \rho 4\pi x^2 dx$$

$$q_{in} = \rho_0 \int_0^r (a - x) 4\pi x^2 dx$$

$$= \rho_0 4\pi \left[ax - \frac{x^3}{3} \right]_0^r$$

$$q_{in} = \frac{\rho_0 4\pi}{3} [3ar - r^2]$$

$$\therefore E(4\pi r^2) = \frac{\rho_0 4\pi}{3\epsilon_0} [3ar - r^2]$$

$$\text{To maximise } E, \frac{dE}{dr} = 0$$

$$\Rightarrow r = \frac{a}{3}$$

4. Ans (A)

Using conservation of energy

$$-\frac{3GM^2}{5R} + \frac{1}{2}MV_0^2 = \frac{-3G(M/8)^2}{5(R/2)} \times 8 + 8 \times \left(\frac{1}{2} \times \frac{M}{8} \times V^2 \right)$$

$$\Rightarrow -\frac{3GM^2}{5R} + \frac{1}{2}MV_0^2 = \frac{-3GM^2}{20R} + \frac{1}{2}MV^2$$

$$\Rightarrow -\frac{3GM^2}{5R} + \frac{GM}{2R} + \frac{3GM}{20R} = \frac{V^2}{2}$$

$$\Rightarrow \frac{GM}{20R} + \frac{V^2}{R}$$

$$\Rightarrow V = \sqrt{\frac{GM}{10R}} = 100 \text{ m/s}$$

PHYSICS

SEC-2

1. Ans (A,B,C,D)

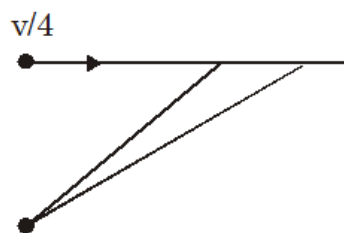
$$\frac{Kq_2^q}{r^2} = m\omega^2 r$$

$$\Rightarrow \frac{9 \times 10^9 \times q^2}{2(\pi^{3/2})^2} = \frac{20}{\sqrt{\pi}} \times 10^{-3} \times \frac{\pi^2}{4} \times \pi^{3/2}$$

$$\Rightarrow q^2 = \frac{40 \times \pi^6 \times 10^{-3}}{9 \times 10^1 \times 4}$$

$$q = \frac{1}{3} \times 100 \times 10^{-6} \text{ C}$$

$$q = \frac{100}{3} \mu\text{C}$$



$$\text{Area} = \pi r^2 \times \frac{t}{4T}, T = \frac{2\pi}{\omega}$$

2. Ans (A,C)

$$\frac{kq}{r} = \frac{V}{2}k \text{ \& } \frac{kq'}{r} = \frac{-V}{2} \Rightarrow q = 2\pi\epsilon_0 rV$$

$$\text{Total charge flows} = q - q' = \frac{r}{k}V$$

$$d = \frac{1}{2} \times q \times \frac{V}{md}t^2$$

$$\frac{2md^2}{2\pi\epsilon_0 rV^2} = t^2 \Rightarrow t = \frac{\sqrt{md}}{\sqrt{\pi\epsilon_0 rV}}$$

$$i_{\text{avg}} = \frac{q}{t} \Rightarrow i_{\text{avg}} = \frac{2(\pi\epsilon_0 r)^{3/2}V^2}{\sqrt{md}}$$

Average power loss =

$$P = \frac{\Delta U}{t} = \frac{q(V/2 + V/2)}{t}$$

$$= \frac{qV}{t} = \frac{(2\pi\epsilon_0 rV)V}{\frac{\sqrt{md}}{\sqrt{\pi\epsilon_0 rV}}} \Rightarrow P \propto V^3$$

3. Ans (B)

We can calculate energy and impulse in method described to observe option (B) & (C) are correct.

Total Energy required for direct projection

$$\frac{1}{2}m\left(\sqrt{\frac{2GM}{R}}\right)^2 = \frac{GMm}{R}$$

Total Energy required for step wise projection

$$\Delta KE_1 = \frac{1}{2}m\left(\sqrt{\frac{GM}{R}}\right)^2$$

$$\Delta KE_2 = \frac{1}{2}m\left(\sqrt{\frac{8}{5}\frac{GM}{R}}\right)^2 - \frac{1}{2}m\left(\sqrt{\frac{GM}{R}}\right)^2$$

$$\Delta KE_3 = \frac{1}{2}m\left(\sqrt{\frac{GM}{2R}}\right)^2 - \frac{1}{2}m\left(\frac{1}{4}\sqrt{\frac{8GM}{5R}}\right)^2$$

$$\Delta KE_{\text{Total}} = \Delta KE_{\text{direct projection}}$$

For Impulse calculation

$$\Delta I_{\text{direct}} = m\sqrt{\frac{2GM}{R}}$$

For step wise projection

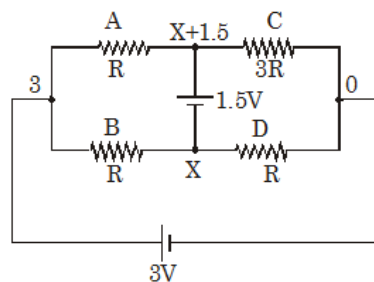
$$\Delta I_1 = m\sqrt{\frac{GM}{R}}$$

$$\Delta I_2 = m\sqrt{\frac{8}{5}\frac{GM}{R}} - m\sqrt{\frac{GM}{R}}$$

$$\Delta I_3 = m\sqrt{\frac{GM}{2R}} - \frac{m}{4}\sqrt{\frac{8}{5}\frac{GM}{R}}$$

$$\Delta I_{\text{direct}} > \Delta I_{\text{stepwise}}$$

4. Ans (A,C)



Applying KCL,

$$\frac{(x-3)}{R} + \frac{(x-0)}{R} + \frac{(x+1.5)}{3R} + \frac{x-1.5}{R} = 0$$

$$10x = 12 \Rightarrow x = 1.5$$

$$\therefore i_A = \frac{0.3A}{R}, i_B = \frac{1.8A}{R}, i_C = \frac{2.7}{3R},$$

$$i_D = \frac{0.2A}{R}$$

5. Ans (C,D)

$$L_i = L_f$$

$$\frac{mR^2}{2}\omega_0 = mvR + \frac{mR^2}{2}\frac{v}{R}$$

$$\frac{mR^2}{2}\omega_0 = \frac{3}{2}mvR$$

$$v = \frac{\omega_0 R}{3}$$

does not depend on μ

$\Rightarrow$ As acceleration depends on μ , time depends on μ .

$\Rightarrow$ work done = ΔKE independent of μ

6. Ans (A,B,C)

Due to earthing, potential of earthed conductor is zero. So, we take three time intervals,

$$(A) t < \frac{a}{v} \quad (B) \frac{a}{v} \leq t < \frac{b}{v} \quad (C) t \geq \frac{b}{v}$$

Let the charge on earth conductor be Q . then

$$(A) \frac{Kq_1}{a} + \frac{Kq_2}{b} + \frac{KQ}{R} = 0 \text{ (both pt. charges are outside)}$$

$$Q = -R\left(\frac{q_1}{a} + \frac{q_2}{b}\right) = -vt\left(\frac{q_1}{a} + \frac{q_2}{b}\right)$$

$$I = \frac{dQ}{dt} = -v\left(\frac{q_1}{a} + \frac{q_2}{b}\right)$$

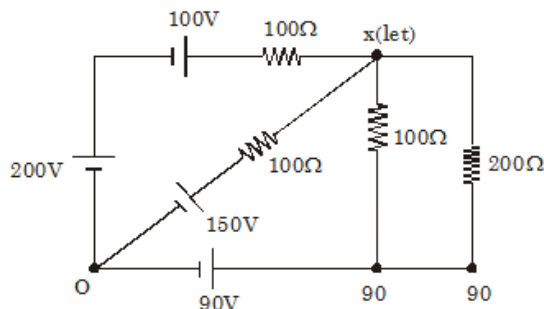
$$(B) \frac{Kq_1}{R} + \frac{KQ}{R} + \frac{Kq_2}{b} = 0 \text{ (} q_1 \text{ is inside \& } q_2 \text{ is outside)}$$

$$(q_1 + Q) = -\frac{Rq_2}{b} = -(vt)\frac{q_2}{b}$$

$$I = \frac{dQ}{dt} = -v\left(\frac{q_2}{b}\right)$$

(C) Finally, when both point charges are inside then ($I = 0$).

8. Ans (A,B,C)



Applying nodal at x i.e. net outward current is zero.

$$\Rightarrow \frac{x-0-300}{100} + \frac{x-0-150}{100} + \frac{x-90}{100} + \frac{x-90}{200} = 0$$

$$\Rightarrow x = \frac{1170}{7}$$

$$\text{Reading of } A_2 = \frac{x-90}{100} = 0.77A$$

$$\text{Reading of voltmeter} = 100 \times 0.7714 = 77.14V$$

$$\text{Reading of } A_1 = 0.77 + 0.385 = 1.16A$$

PHYSICS

SEC-3

1. Ans (3.00)

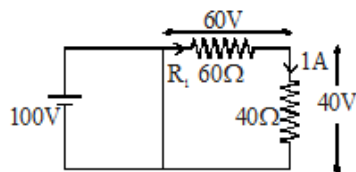
Orbital velocity = 2 m/s

$$\sqrt{\frac{GM}{R}} = 2$$

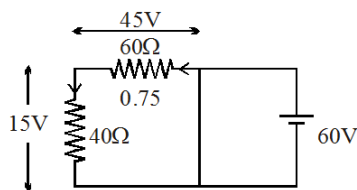
$$R = \frac{GM}{4} = \frac{\frac{20}{3} \times 10^{-11} \times 500\pi \times \frac{4}{3}\pi R^3}{4}$$

On solving $R = 3.00 \text{ km}$

2. Ans (60.00)



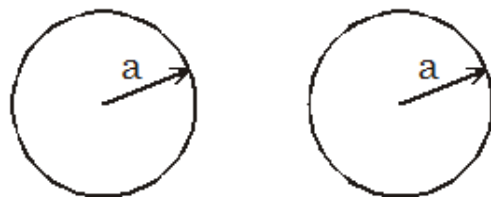
$$R_3 = 40 \Omega$$



$$R_1 = \frac{15}{3/4} = 20\Omega$$

$$\frac{R_1 + R_3}{10} = 6$$

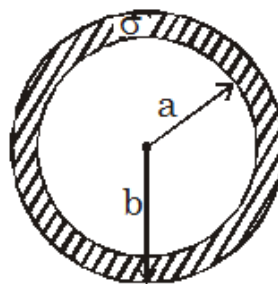
3. Ans (4.75 to 4.80)



As the two spheres are far apart we need to calculate resistance between them

Lets take two concentric shell of radius 'a' and 'b' and space between them is filled by material of conductivity ' σ '

$$R = \frac{1}{4\pi\sigma} \left(\frac{1}{b} - \frac{1}{a} \right)$$



$$\text{If } b \gg a \Rightarrow R = \frac{1}{4\pi\sigma a}$$

Using the previously obtained result, we note

that essentially all of the resistance is in the region right around the inner sphere (radius a).

Successive shells as you go out contribute less

and less, because the cross-sectional area ($4\pi r^2$)

gets larger and larger. For the two submerged

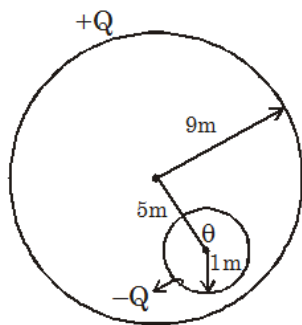
spheres $R = \frac{2}{4\pi\sigma a}$ (on R as the current leaves

the first, one R as it converges on the second).

Therefore

$$I = \frac{V}{R} = 2\pi\sigma aV$$

4. Ans (11.00)



Self energy of spheres

$$= \frac{k(Q)^2}{2(9)} + \frac{k(-Q)^2}{2(1)}$$

Interaction energy

$$= \frac{k(Q)(-Q)}{1} + \frac{k(Q)(Q)}{9} + \frac{k(Q)(-Q)}{9}$$

$$= -\frac{kQ^2}{1}$$

Total final potential energy

$$= \frac{kQ^2}{18} + \frac{kQ^2}{2} - \frac{kQ^2}{1} = -\frac{8}{18}(kQ^2)$$

$$= -16 \times 10^3$$

$$V_f - V_i = -16 \times 10^3$$

[As potential energy at infinity = 0]

$$\Rightarrow V_i = 0$$

$$\text{Work done} = -\Delta U = 16 \times 10^3$$

comparing

$$\alpha = 0, 2\beta = 16, \gamma = 3$$

$$\alpha + \beta + \gamma = 11$$

5. Ans (5.00)

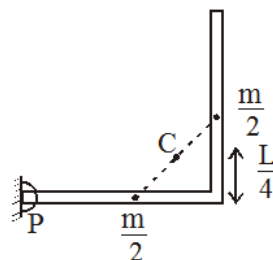
$$I_p = \frac{1}{3} \frac{m}{2} L^2 +$$

$$\left[\frac{1}{12} \frac{m}{2} L^2 + \frac{m}{2} \left(\frac{L^2}{4} + L^2 \right) \right]$$

$$= \frac{5}{6} mL^2$$

' ω ' will be max. when COM is at its lowest position.

$$\therefore \text{vertical distance moved by COM} = \frac{\sqrt{10}L}{4} + \frac{L}{4}$$



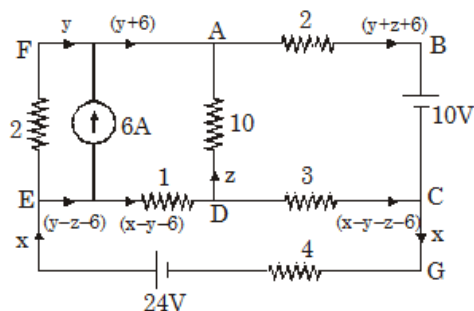
$$PC = \frac{\sqrt{10}L}{4}$$

$$mg \left[\sqrt{10} + 1 \right] \frac{L}{4} = \frac{1}{2} I_p \omega^2 = \frac{5}{12} mL^2 \omega^2$$

$$\Rightarrow \omega^2 = \frac{3g}{5L} (\sqrt{10} + 1) \therefore \omega = 5 \frac{r}{s}$$

6. Ans (4.05 to 4.35)

The given circuit is redrawn with assumed distribution of currents in figure. Applying KVL to different closed loops, we get



Circuit EFADE

$$-2y + 10z + (x - y - 6) = 0 \text{ or } x - 3y + 10z = 6$$

Circuit ABCDA

$$2(y + z + 6) - 10 + 3(x - y - z - 6) - 10z = 0$$

$$\text{or } 3x - 5y - 14z = 40$$

Circuit EDCGE

$$-(x - y - 6) - 3(x - y - z - 6) - 4x + 24 = 0 \text{ or}$$

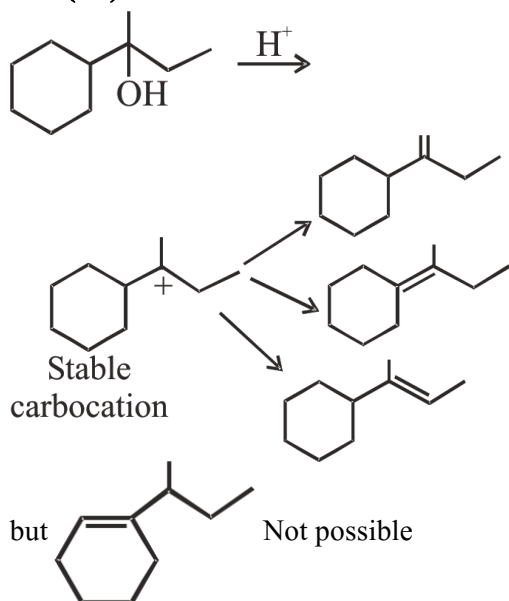
$$8x - 4y - 3z = 48$$

From above equations we get $x = 4.1 \text{ A}$

CHEMISTRY

SEC-1

2. Ans (B)



3. Ans (B)

$$A_t = \frac{A_0}{2^3} = \frac{10^7}{8} \text{ dps} \Rightarrow \frac{dN_\alpha}{dt} = 2 A_t = \frac{10^7}{4} = 2.5 \times 10^6 \text{ s}^{-1}$$

4. Ans (D)

according to slow step

$$r(t) = K_2 [N_2O_2] [H_2]$$

$$\text{Where } K_1 = \frac{[N_2O_2]}{[NO]^2}$$

$$[N_2O_2] = K_1 [NO]^2$$

$$\text{So, } r(t) = K_2 K_1 [NO]^2 [H_2]$$

$$\text{So, } r(t) = K_2 K_1 [NO]^2 [H_2] K.$$

$$-\frac{1}{2} \frac{d[H_2]}{dt} = -\frac{1}{2} \frac{d[NO]}{dt} = \frac{d[N_2]}{dt} =$$

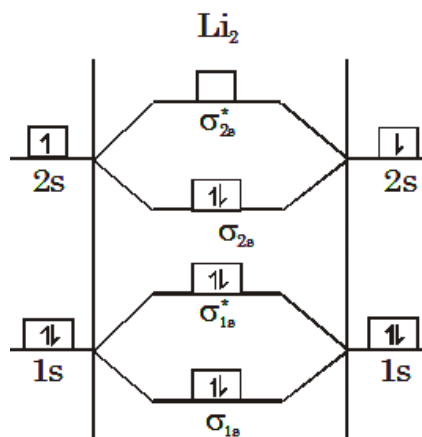
$$K [NO]^2 [H_2]$$

$$\text{Where } K = K_1 K_2$$

CHEMISTRY

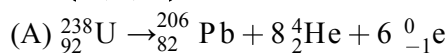
SEC-2

3. Ans (A,B,C)



$$\sigma_{2s}^* \left(\begin{array}{c} + \\ \cdot \end{array} \quad \begin{array}{c} \cdot \\ - \end{array} \right) \text{ M.O. is vacant in ground state}$$

6. Ans (A,C,D)



Now, 89.6 ml at NTP $\Rightarrow$ 4 m mole He

$$\Rightarrow n_{\text{Pb}} = \frac{1}{8} n_{\text{He}} = \frac{1}{8} \times 4 = \frac{1}{2} \text{ m mole}$$

$$\therefore \text{mass of Pb} = 206 \times \frac{1}{2} \text{ mg} = 103 \text{ mg}$$

(B) For radioactive decay rate is independent of temperature.

(C), (D) factual

7. Ans (A,B,C,D)

Distance between nearest neighbour = $\frac{\sqrt{3}a}{2}$ (= 8 in number). Distance between next nearest neighbour = a (= 6 in number)

CHEMISTRY

SEC-3

4. Ans (14.00)

$$K_I = \frac{\ln 2}{t_{1/2}} = \frac{0.7}{t_{1/2}}$$

$$K_{II} = \frac{0.1}{2 \times t_{1/2}}$$

$$\frac{K_I}{K_{II}} = 14$$

MATHEMATICS

SEC-1

1. **Ans (B)**

$\therefore$ on differentiation $(1 - x^2)y_1^2 = a^2 y^2$
 on differentiation $(1 - x^2)y_3 - (a^2 + 1)y_1 = 3xy_2$
 so we get on differentiation
 $(1 - x^2)y_{n+2} - (n^2 + a^2)y_n = (2n + 1)xy_{n+1}$

2. **Ans (C)**

$\therefore \cos 5x$ is discontinuous at
 $5x = \frac{\pi}{2}, \frac{3\pi}{2}, \dots, \frac{9\pi}{2}$
 $g(x) = \{m \sin x\}$ is non-derivable at $(2m - 1)$
 points $2m - 1 = 5 \Rightarrow m = 3$

3. **Ans (B)**

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{e^{3x} f(x)}{e^{3x}} = \lim_{x \rightarrow \infty} \frac{f'(x) + 3f(x)}{3}$$

$$= \frac{12}{3} = 4$$

4. **Ans (A)**

$f(x) + f(y) = f(x)f(y) + f(xy)$
 partial differentiation wrt x
 $f'(x) = f'(x)f(x) + f'(xy) \cdot (y)$
 put $x = 1$
 $-2 = -2f(y) + f'(y)y$
 get $f(y) = 1 - y^2 \Rightarrow f(x) = 1 - x^2$
 Now, number of solution(s) of $f(x) = 2$
 $1 - x^2 = 2 \Rightarrow x^2 = -1 \Rightarrow x \in \phi$

MATHEMATICS

SEC-2

1. **Ans (B)**

Slopes of tangents are given by $y' = \frac{x_1}{y_1} = m_1$
 and $y' = -\frac{x_1}{y_1} = m_2$
 $\tan \theta = \left| \frac{\frac{x_1}{y_1} + \frac{x_1}{y_1}}{1 - \frac{x_1^2}{y_1^2}} \right| = \left| \frac{2x_1 y_1}{a^2} \right| = 1 \Rightarrow \theta = \frac{\pi}{4}$

2. **Ans (A,B,C,D)**

Let $\lambda^{2017} = n$
 now $(\sqrt{n} + \sqrt{n+1}) \in (\sqrt{4n+1}, \sqrt{4n+3})$
 $\dots(1)$
 now as $4n+2$ & $4n+3$
 can never be perfect squares
 $\Rightarrow [\sqrt{4n+2}] = [\sqrt{4n+3}] = k \quad (\text{let}) \dots$
 (2)
 Using (1) & (2)
 $[\sqrt{n} + \sqrt{n+1}] = [\sqrt{4n+1}] = [\sqrt{4n+2}]$
 $= [\sqrt{4n+3}] = \left[\frac{1}{\sqrt{n+1} - \sqrt{n}} \right]$

3. **Ans (A,B)**

$\therefore \lim_{x \rightarrow \infty} \frac{f(x)}{|x|^3} = 0 \Rightarrow f(x)$ is either linear or quadratic
 Let $f(x) = ax^2 + bx$ ($\because f(0) = 0$)
 $\therefore$
 $\lim_{x \rightarrow \infty} (\sqrt{f(x)} - x) = \lim_{x \rightarrow \infty} \frac{(a-1)x^2 + bx}{x \left(\sqrt{a + \frac{b}{x}} + 1 \right)} = -1$
 $\Rightarrow a - 1 = 0$ and $\frac{b}{\sqrt{a} + 1} = -1 \Rightarrow b = -2$
 Hence, $f(x) = x^2 - 2x$
 Now proceed.

4. Ans (A,C)

$$(1 + \sqrt{5})^n = \alpha_n + \beta_n \sqrt{5} \quad \dots(1)$$

$$(1 - \sqrt{5})^n = \alpha_n - \beta_n \sqrt{5} \quad \dots(2)$$

equation (1) and (2)

$$\Rightarrow (-4)^n = \alpha_n^2 - 5\beta_n^2 \Rightarrow$$

$$(-1)^n = \left(\frac{\alpha_n}{2^n}\right)^2 - 5\left(\frac{\beta_n}{2^n}\right)^2$$

Now put $n \rightarrow n+1$ in equation (1)

$$\Rightarrow \alpha_{n+1} + \beta_{n+1} \sqrt{5} = (1 + \sqrt{5})^{n+1} =$$

$$(1 + \sqrt{5})(1 + \sqrt{5})^n = (1 + \sqrt{5})(\alpha_n + \beta_n \sqrt{5})$$

$\Rightarrow$

$$\alpha_{n+1} + \beta_{n+1} \sqrt{5} = \alpha_n + \beta_n \sqrt{5} + \sqrt{5}\alpha_n + 5\beta_n$$

$$\Rightarrow \alpha_{n+1} = \alpha_n + 5\beta_n \quad \dots(3)$$

$$\Rightarrow \beta_{n+1} = \beta_n + \alpha_n \quad \dots(4)$$

from equation (3) and (4) we have

$$\Rightarrow \alpha_{n+1} - \beta_{n+1} = \alpha_n + 5\beta_n - \beta_n - \alpha_n - \beta_n - \alpha_n$$

$$\Rightarrow \frac{\alpha_{n+1} - \beta_{n+1}}{3} = \frac{4}{3}\beta_n, \text{ now check the}$$

option(s).

$$\lim_{\theta \rightarrow 0} \frac{\tan^{-1}\left(\frac{4}{3}\beta_n\theta\right)}{\sin(\beta_n\theta)} = \frac{4}{3}$$

5. Ans (C)

Substituting $y = a \sin \theta$

$$x = \frac{a}{2} \log \left(\frac{a + a \cos \theta}{a - a \cos \theta} \right) - a \cos \theta$$

$$= a \log \left(\cot \frac{\theta}{2} \right) - a \cos \theta$$

$$\therefore \frac{dx}{d\theta} = -\frac{a \cos^2 \theta}{\sin \theta}$$

$$\text{also } \frac{dy}{d\theta} = a \cos \theta$$

$$\therefore \frac{dy}{dx} = \frac{a \cos \theta \sin \theta}{-a \cos^2 \theta} = -\tan \theta$$

equation of tangent $Y - y = -\tan \theta (X - x)$

$$Y - y = -\frac{y}{\sqrt{a^2 - y^2}} (X - x)$$

$$Y = 0; X = x + \sqrt{a^2 - y^2} = x_1 \text{ (say)}$$

$$PT^2 = a^2 - y^2 + y^2 \Rightarrow PT = a = \text{constant}$$

6. Ans (B,C)

$$f(x^2) - xf(x) = x^4(x^2 - 1)$$

$$\Rightarrow xf(x^2) - x^2f(x) = x \cdot x^3 \left((x^2)^2 - x^2 \right)$$

$$\Rightarrow \frac{f(x^2)}{x^2} - \frac{f(x)}{x} = (x^2)^2 - x^2$$

$$\Rightarrow \frac{f(x)}{x} - x^2 = k \Rightarrow f(x) = x^3 + kx$$

7. Ans (B,C,D)

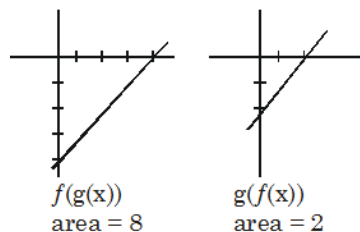
$$f(x) = \begin{cases} -4 - 2x & x \in [-1, 0) \\ -2x & x \in [0, 1) \\ 4 - 2x & x \in [1, 2) \\ 8 - 2x & x \in [2, 3) \end{cases}$$

$$g(x) = \begin{cases} -2 - \frac{x}{2} & x \in [-2, 0) \\ -\frac{x}{2} & x \in [0, 2) \\ 2 - \frac{x}{2} & x \in [2, 4) \\ 4 - \frac{x}{2} & x \in [4, 6) \end{cases}$$

$f(x)$ and $g(x)$ both are bijective

$$f(g(x)) = x - 4 \quad \forall x/I$$

$$g(f(x)) = x - 2 \quad \forall x/I$$



8. Ans (B,D)

$$f(f(0)) = f(2) = 8 - 4 + 2 = 6$$

$$f(f(2)) = f(6) = 3 - 2(6) = -3$$

$f(f(x))$ is discontinuous at 3 points

MATHEMATICS

SEC-3

1. Ans (1.75)

$$f(2x) = f(x) =$$

$$f\left(\frac{x}{2}\right) = f\left(\frac{x}{2^2}\right) \dots \dots = f\left(\frac{x}{2^n}\right)$$

$$\text{if } \lim_{n \rightarrow \infty} f\left(\frac{x}{2^n}\right) = f(0) = f(2016) = \frac{\pi}{2}$$

$$\lim_{x \rightarrow 0} \frac{\cos \frac{\pi}{2} - \sqrt[3]{\cos \frac{\pi}{2}}}{\sin^2 x} = 0 = a$$

and $g(x)$ is non-differentiable

$$-\frac{3\pi}{4}, -\frac{\pi}{2}, -\frac{\pi}{4}, 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}$$

$$\text{So, } b = 7$$

2. Ans (7.50)

$$\lim_{x \rightarrow 0} \frac{(\cos 6x)^{11} (\cos 4x)^9 - (\cos 2x)^3 (\cos 5x)^5 (\cos 7x)^7}{x^2}$$

$$\lim_{x \rightarrow 0} \frac{(1 - 18x^2)^{11} (1 - 8x^2)^9 - (1 - 2x^2)^3 \left(1 - \frac{25x^2}{2}\right)^5 \left(1 - \frac{49x^2}{2}\right)^7}{x^2}$$

$$\Rightarrow L = -30$$

3. Ans (0.25)

$$\lim_{x \rightarrow 0} \frac{1}{x^3} \left(\frac{(1 - bx) - (1 - ax)\sqrt{1+x}}{\sqrt{1+x}(1 - bx)} \right)$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{1}{x^3} \left[\frac{(1 - bx) - (1 - ax)(1 + \frac{x}{2} - \frac{1}{8}x^2 + \frac{1}{16}x^3 + \dots)}{\sqrt{1+x}(1 - bx)} \right]$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{(a - b - \frac{1}{2})x + (\frac{a}{2} + \frac{1}{8})x^2 + (-\frac{1}{16} - \frac{a}{8})x^3 + \dots}{x^3 \sqrt{1+x}(1 - bx)}$$

$$\Rightarrow a - b - \frac{1}{2} = 0 \quad \& \quad \frac{a}{2} + \frac{1}{8} = 0 \quad \text{and}$$

$$\ell = -\frac{1}{16} - \frac{a}{8}$$

$$\Rightarrow a - b = \frac{1}{2} \quad \& \quad a = -\frac{1}{4}$$

$$b = -\frac{3}{4} \quad \& \quad a = -\frac{1}{4} \quad \text{and}$$

$$\ell = -\frac{1}{16} + \frac{1}{32} = -\frac{1}{32}$$

$$\Rightarrow \frac{1}{a} - \frac{3}{b} - 32\ell = -4 + 4 + 1 = 1$$

4. Ans (0.40)

Let $y = mx + c$ is a bitangent, therefore the equation $x^4 + 2x^3 - 11x^2 - 13x + 35 = mx + c$ must have exactly two real solutions.

therefore the expression $x^4 + 2x^3 - 11x^2 - (13+m)x + (35-c)$ is the product of the squares of two linear factors, so perfect square of a quadratic expression.

$$\text{Let } x^4 + 2x^3 - 11x^2 - (13+m)x + (35-c) = (x^2 + px + q)^2$$

$$\text{comparing coefficient of } x^3, 2p = 2 \Rightarrow p = 1$$

$$\text{comparing coefficient of } x^2, p^2 + 2q = -11$$

$$\Rightarrow q = -6$$

$$\therefore (x^2 + px + q)^2 = (x^2 + x - 6)^2$$

$$\therefore m = c = 1 \quad \therefore y = -x - 1 \text{ is bitangent}$$

$$\therefore |m + c| = 2$$

5. Ans (0.00)

Since, n is even, let $n = 2\ell$

Now,

$$\begin{aligned} & 1 + 2x + 3x^2 + 4x^3 + 5x^4 + 6x^5 + 7x^6 + 8x^7 + \dots + (2\ell + 1)x^{2\ell} \\ &= (1 + 2x + x^2) + 2x^2(1 + 2x + x^2) + 3x^4(1 + 2x + x^2) \\ &+ 4x^6(1 + 2x + x^2) + \dots + \ell \cdot x^{2\ell-2}(1 + 2x + x^2) + (\ell + 1)x^{2\ell} \\ &= (1 + x)^2(1 + 2x^2 + 3x^4 + 4x^6 + \dots + \ell \cdot x^{2\ell-2}) + (\ell + 1)x^{2\ell} \end{aligned}$$

Which can never be zero for any value of x .

6. Ans (0.00)

$$\lim_{x \rightarrow \frac{\pi}{4}} \frac{4\sqrt{2} - (\cos x + \sin x)^5 - 10 \cos x(1 - \sin 2x)}{1 - \sin 2x}$$

$$= \lim_{x \rightarrow \frac{\pi}{4}} \left(\frac{4\sqrt{2} - (\sqrt{2} \cos(x - \frac{\pi}{4}))^5}{1 - \sin 2x} \right)$$

$$- \lim_{x \rightarrow \frac{\pi}{4}} 10 \cos x$$

$$= \lim_{x \rightarrow \frac{\pi}{4}} 4\sqrt{2} \left(\frac{1 - \cos^5(x - \frac{\pi}{4})}{1 - \sin 2x} \right) - \frac{10}{\sqrt{2}}$$

$$= \lim_{h \rightarrow 0} \frac{4\sqrt{2}(1 - \cos^5 h)}{1 - \cos 2h} - 5\sqrt{2}$$

$$= \lim_{h \rightarrow 0} \frac{4\sqrt{2}(1 - \cos^5 h)}{2(1 - \cos 2h)} - 5\sqrt{2}$$

$$= \lim_{h \rightarrow 0} \frac{4\sqrt{2}(1 + \cos h + \cos^2 h + \cos^3 h + \cos^4 h)}{2(1 + \cosh)} - 5\sqrt{2}$$

$$= 5\sqrt{2} - 5\sqrt{2} = 0$$